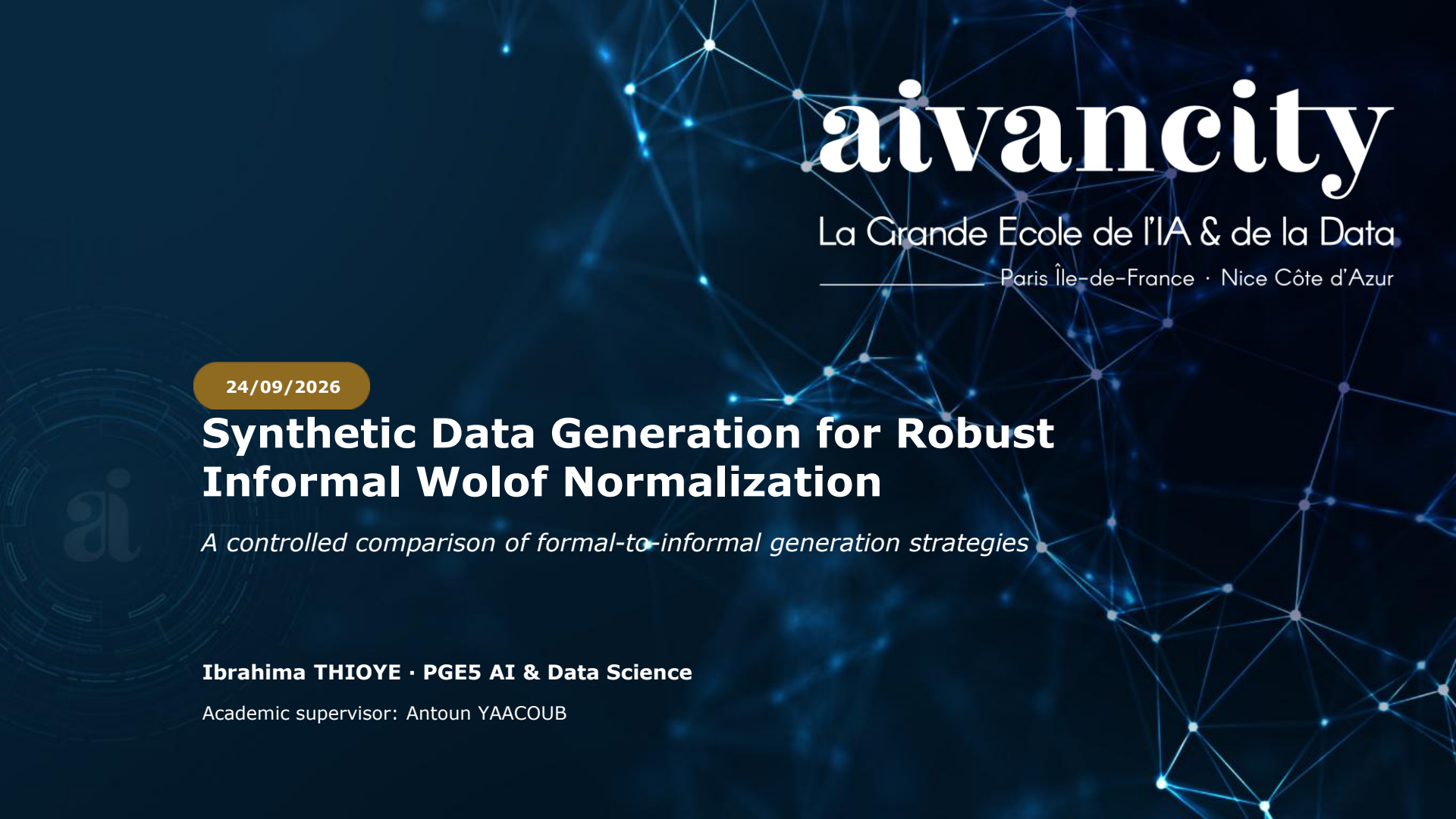




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La Grande Ecole de l'IA & de la Data

Paris Île-de-France · Nice Côte d'Azur

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Synthetic Data Generation for Robust Informal Wolof Normalization

A controlled comparison of formal-to-informal generation strategies

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Informality and code-switching can alter translation

INFORMAL CODE SWITCHED INPUT

mareme thiam à la fin lanegua beugue

FORMAL INPUT

mareme thiam à la fin lan nga bëgg

TARGET TRANSLATION

mareme thiam au final, que veux-tu ?

GOOGLE TRANSLATE

FORMAL → FRENCH

mareme thiam au final, que veux-tu ?

INFORMAL CODE-SWITCHED → FRENCH

marée thiam à la fin lanegua veut

DEEPL

FORMAL → FRENCH

Mareme Thiam, au fond, qu'est-ce que tu veux ?

INFORMAL CODE-SWITCHED → FRENCH

Mareme Thiam, à la fin, la phrase est incohérente.

Same task and meaning; spelling and language mixing change the system response.

The two closest precedents

BEQI

Wolof spelling correction combining linguistic knowledge and neural modeling. It motivates M1's transparent Wolof-specific orthographic rules.

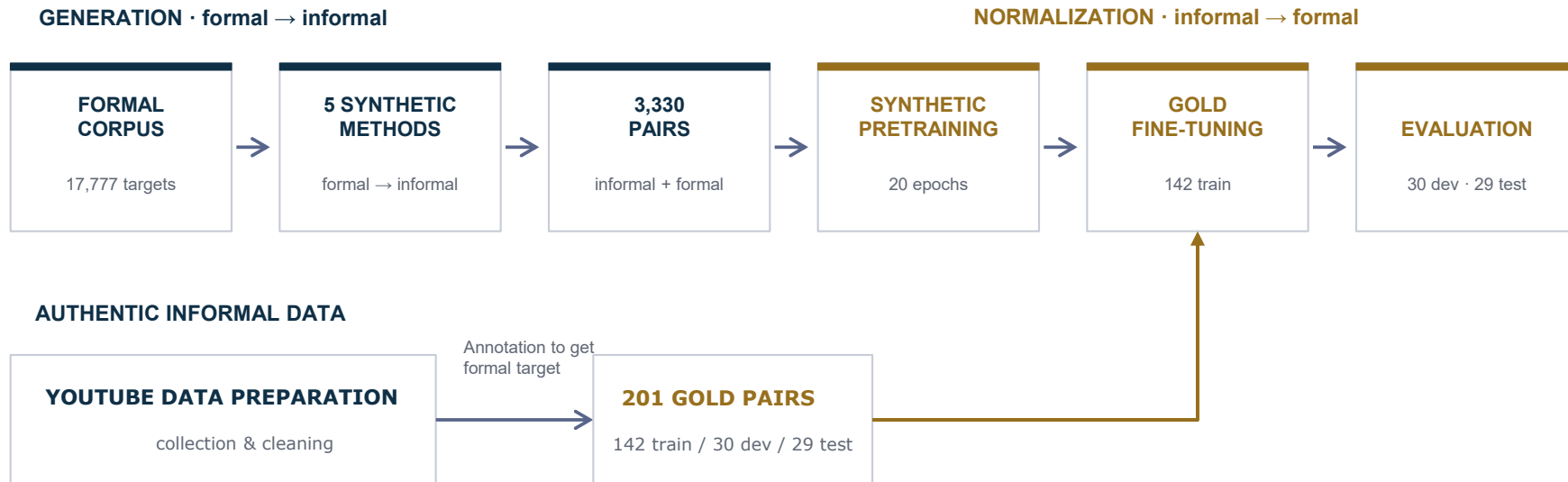
OOLEL

A published formal/non-standard Wolof dataset generated with a Wolof-adapted LLM. It enters the benchmark directly as external method M4.

Research question and project position

How well do formal-to-informal synthetic generation strategies improve normalization of authentic, code-switched Wolof YouTube comments under the same architecture?

The benchmark reverses the generation direction during training



Why 3330 pairs ? Because 3430 pairs were found exactly in the 17777 dataset

From raw YouTube comments to a frozen benchmark



ANNOTATED GOLD PAIR · source_index 5879

201 GOLD PAIRS

Informal: doff bu reww → Formal doff bu reew / decision: KEEP

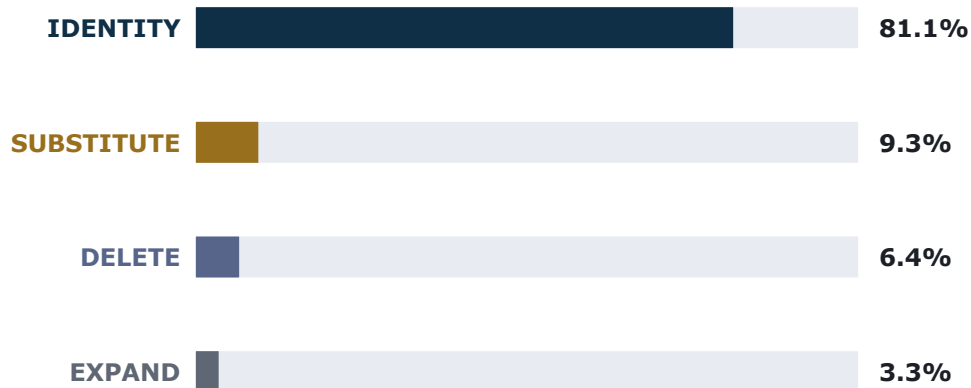
ENRICHED TOKEN · token_position 0

1,315 OCCURRENCES

doff → dof / lemma: dof / POS: NOUN / language: wo→wo / entity: UNKNOWN / protected: NO · error: REPETITION

Identity is the dominant gold transformation

GOLD-TRAIN OPERATION SHARE



Identity means KEEP: the source character is preserved.

Realistic noise requires many preserved positions and a small number of targeted edits.

CODE-SWITCHING IN GOLD

62.0% of sentences contain French.

PROTECTION

31.0% of tokens are protected

GRAMMATICAL CONTEXT

Change rate by POS : verbs 77.2%, pronouns 57.1%, nouns 48.9%, determiners 24.8%.

Five synthetic generation methods

M1 FAMILY

M1 RULES

Beqi-inspired
regex rules

M2 ALIGNMENT

French substitutions
then M1

M3 CMDR

Code Mixing from
Distributional
Representations
then M1

INDEPENDENT METHODS

M4 OOLEL

External LLM
pairs

M6 PROPOSED

Gold-informed
engine

CONTROLS

Identity: return the input unchanged · Gold-only: train without synthetic pretraining

M1 supplies spelling noise; M2 adds aligned French

M1 · DETERMINISTIC RULES

Mask entities → apply 27 ordered regex rules → restore protected spans.

Rule examples:

ñ→gn · ë→eu · u→ou
x/q→kh · aa/ee/oo→a/e/o

Nit ñi wáy di way.

→ nit gni wáy di way.

Trace: lowercase + ñ→gn

M2 · ALIGNMENT + M1

Eflomal probabilities propose French words found in the paired French sentence; target ≈20% of eligible positions; protect insertion; apply M1.

Nit ñi wáy di way.

→ nit les wáy di way.

Trace: ñi→les + M1

M1 risk: broad frequency mismatch

M2 risk: alignment ≠ contextual validity

M3 mixes by similarity; M4 imports LLM variation

M3 · CMDR + M1

- Shared French-Wolof Word2Vec space
- Score = similarity x bilingual evidence x position x n-gram penalty.
- apply M1.
-

Joseph: Déet waay, Jasmine.
→ jasmine: det way, joseph.

M3: vector proximity can swap entities

M4 · OOEL

Import 3,433 valid standard/non-standard pairs. All 3,430 unique formal targets overlap the formal pool.

Joseph: Déet waay, Jasmine.
→ Joseph: Déet waye Jasmine

M4: useful data, opaque generation settings

M6 learns selective transformations from gold train

1 FILTER

Keep reviewed, eligible Wolof occurrences; protect French and entities.

2 MAP

Learn 171 dominant formal→informal lexical mappings.

3 GENERALIZE

Learn 105 POS/context-conditioned character templates.

4 SAMPLE

Choose a deterministic edit budget; reject overlaps.

RECORDED M6 TRACE · exported row parallel:11883

1 · FORMAL INPUT

na nga def
Eligible Wolof span



2 · LEXICAL MAP

na nga → nanga
spacing_merge + phonetic
1 gold occurrence



3 · CONTEXT RULE

def → deff
VERB · suffix repetition
3 gold occurrences



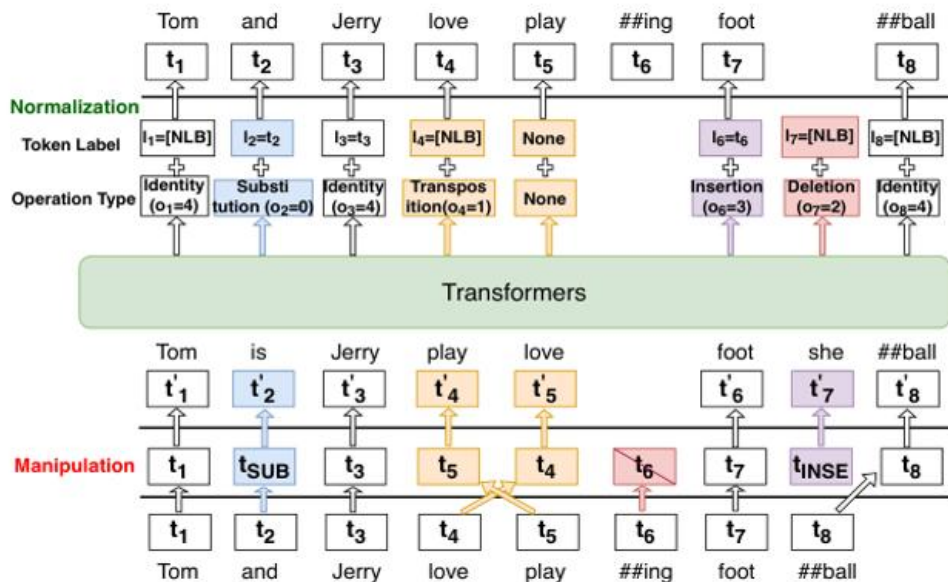
4 · OUTPUT

nanga deff
2 non-overlapping edits

The sampler combines one dominant mapping with one POS/context template, then rejects overlaps.

Gold-train supervision only · no M1 rules · no development/test annotations

TNT (Text Normalization based pre-training of Transformers) as the evaluator



WHY THIS MODEL?

- Character-level input handles unseen spellings.
- Predict KEEP / DELETE / SUBSTITUTE / EXPAND.
- 2.15M parameters: feasible for repeated conditions.
- No autoregressive decoding or exposure bias.

No single metric captures a good correction

CER ↓

Character edit distance. Sensitive to partial spelling repair.

WER ↓

Whole-word error. A partly repaired token is still wrong.

chrF ↑

Character n-gram overlap. Graded surface similarity.

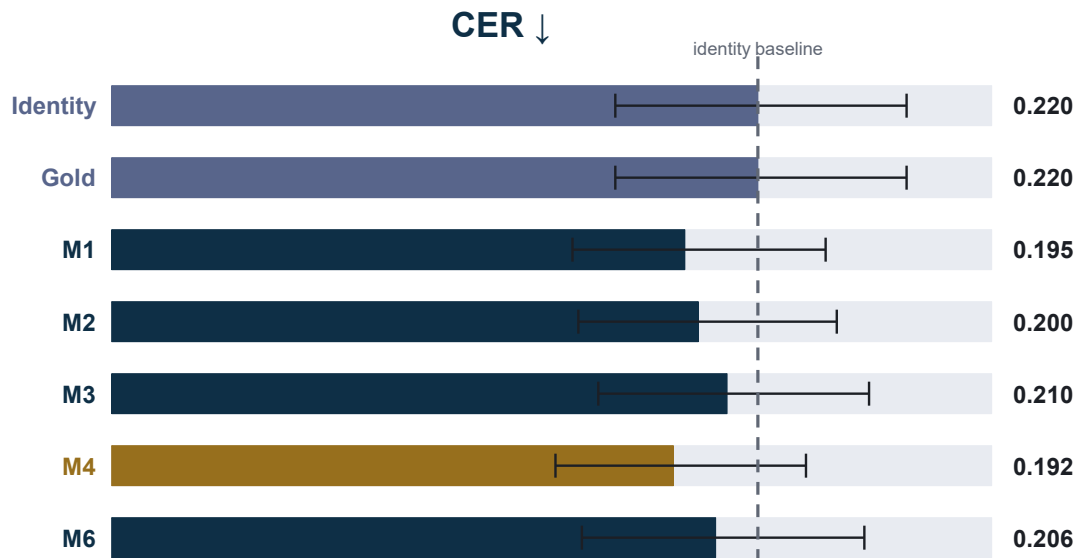
Overcorrection ↓

False proposed edits / all proposed edits.

Exact sentence ↑

Entire prediction must match the reference.

M4 and M1 lead on character-level transfer



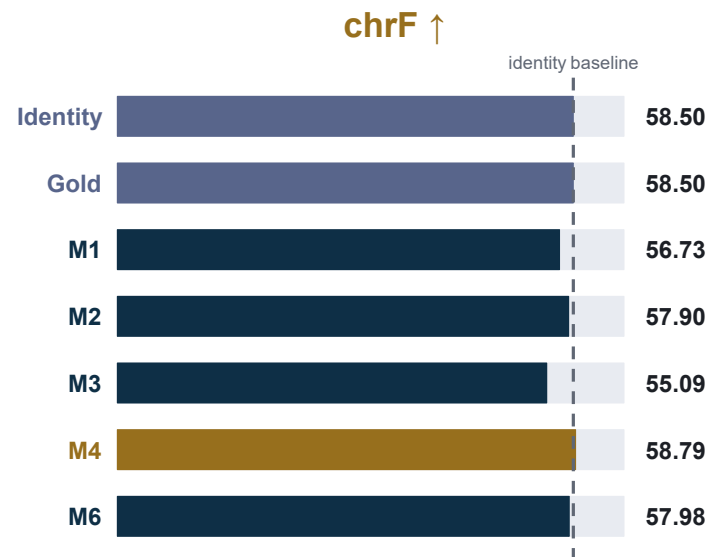
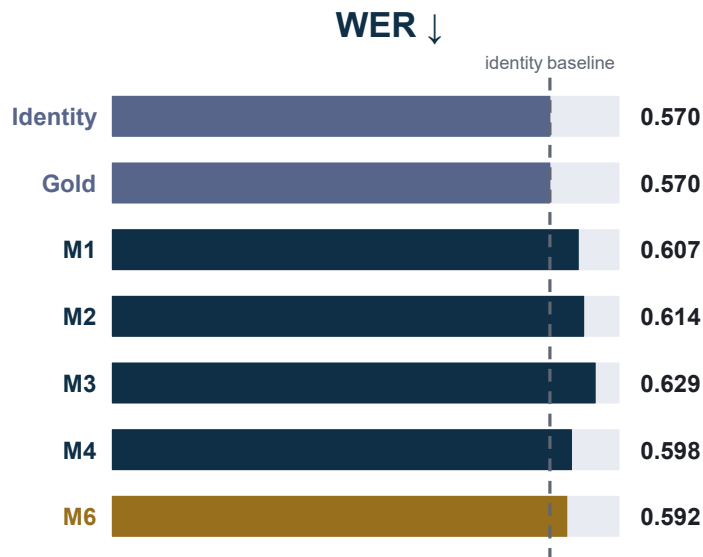
M4 OOLEL

CER 0.191
13.1% below identity

M1 RULES

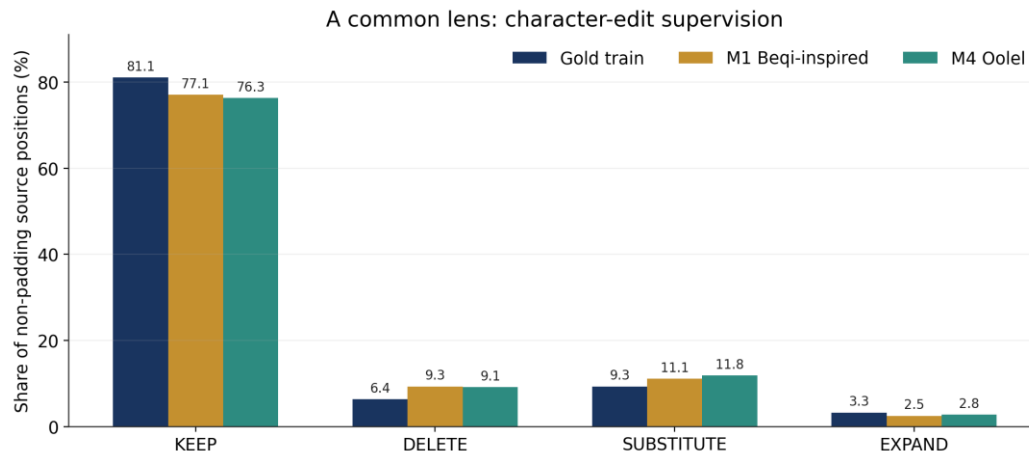
CER 0.195
11.3% below identity

Character gains do not become reliable word recovery



Best synthetic WER: M6 · only trained condition above identity on chrF: M4 (+0.29)

Edit density explains M1/M4 gains and M6's shortfall



WHY M6 LOSES CER

95.6% KEEP

Gold train: 81.1% KEEP

SUBSTITUTE 1.25% · Gold 9.32%

EXPAND 0.69% · Gold 3.26%

Spacing merges: 0.4% of rows

WER 0.592 · best synthetic
CER 0.206 · fourth of five

M6 copies safely, but under-exposes the edits needed to lower CER.

Overcorrection reveals calibration failures

False exact edits / proposed edits



M6: lowest overcorrection (96.5%); F1 = 0.030.

EXACT SENTENCE FAILURE

M1: 28/29 failed — **96.6%**
M2: 29/29 failed — **100%**
M3: 29/29 failed — **100%**
M4: 29/29 failed — **100%**
M6: 29/29 failed — **100%**

M2 · BROAD INSERTION

French inserted into 3,148/3,330 rows (94.5%; 2.62/row). CER and WER do not beat M1.

M3 · SELECTIVE BUT UNSAFE

French inserted into 1,094/3,330 rows (32.9%; 0.46/row). Similarity can swap invalid words or entities.

More injected French did not produce consistent gains.

What the pilot establishes - and what it cannot

LIMITATIONS

- Too little data: 201 pairs from five videos
- No parameter calibration after gold annotation (except M6)
- Small character model, limited semantics
- Only one annotator
- Only tested on normalization

NEXT STEPS

- broaden channels, topics, data source
- calibrate synthetic edit density on gold train
- test stronger/multiple evaluators
- Involve more annotators
- measure retrieval and translation gains

Synthetic data helps - but calibration decides how

Synthetic pretraining transfers useful character patterns; none of the tested conditions delivers reliable word-level or exact normalization.

1 · AUTHENTIC BENCHMARK

A manually normalized train/dev/test corpus split by video.

2 · CONTROLLED COMPARISON

Five generators, matched formal targets, one evaluator and one protocol.

3 · AUDITABLE ANALYSIS

Gold transformations, generator mechanisms and strict edit behavior.

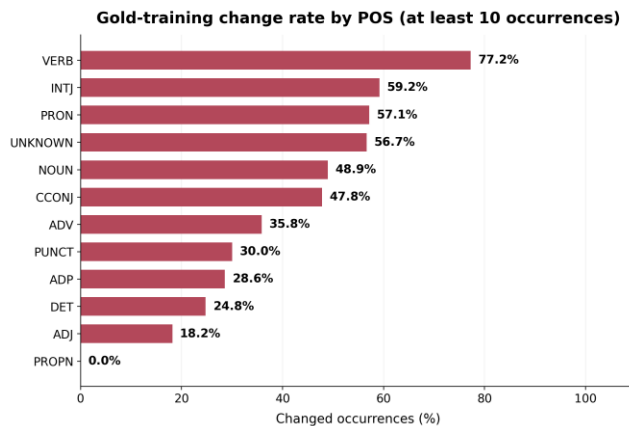
Thank you · Questions

Backup slides

Additional evidence for discussion

Gold patterns · generator mechanics · frozen predictions · reproducibility

Change rates differ sharply by grammatical category



Frozen predictions reveal partial repair and accumulated errors

SHORTER EXAMPLE

Input: mareme mom dey forcer trop vraiment amoufi place
Gold: mareme moom dey forcer trop vraiment amu fi place
M1/M4/M6: mareme mom dey forcer trop vraiment amufi place

Character repair occurs, but vowel length and spacing remain wrong.

LONGER EXAMPLE

Input: bass miii kou beugue leuffff leu boy biii...
M4: bass miii ku bēgēe lēffff lu boy biii...
M6: bass miii kñu bggggg léfff lē boy bijj...

Locally plausible edits accumulate into new corruption.

These are frozen-test normalizer outputs - not generator examples.

Reproducibility and comparison boundaries

LOCKED

Video split · seed 2026 · 3,330 target IDs · model architecture · schedules · metric code · frozen test

AUDITABLE

Generator traces · source IDs · M6 evidence counts · prediction files · training plots · manifests

NOT REPRODUCIBLE

Oolel's original LLM prompt, sampling seed and decoding configuration are not available in this project.

The benchmark evaluates saved data utility; it does not claim identical generation-cost reproducibility.

Where the formal and authentic data come from

FORMAL REFERENCE POOL

17,777 French-Wolof pairs
Hugging Face: galsenai/french-wolof-translation

Google SMOL	7,405
mafand	6,366
FLORES	2,009
Microsoft NTREX	1,997

WHAT 'FORMAL' MEANS HERE

The Wolof side is used as the operational standardized target of curated translation resources.

Checks: Unicode/empty values, canonicalization, duplicates, common targets, hashes.

Limit: all 17,777 rows were not independently re-certified by a Wolof linguist.



What was manual, automatic, and scored?

MANUAL GOLD ANNOTATION

Complete informal-to-formal sentence
Token correspondences
POS, language, entity, protection
Multi-label linguistic errors

201 pairs; enrichment only on 142 Gold-train sentences

AUTOMATIC TNT TARGETS

Levenshtein character alignment

KEEP same character
DELETE emit nothing
SUBSTITUTE one new character
EXPAND multi-character segment

Gold train: 81.06% KEEP

AUTOMATIC EVALUATION

CER character edit distance
WER token edit distance
chrF character n-gram F-score
Exact whole sentence

Correction P/R/F1 uses exact source-span + replacement signatures

String operation is not linguistic interpretation or an evaluation signature

M2 and M3: two code-switching mechanisms

M2 | EFLOMAL ALIGNMENT

Learn Wolof-to-French probabilities from 17,777 paired sentences.

Keep links: count ≥ 3 and probability ≥ 0.10 .

Replace about 20% of eligible positions with French words from the paired French sentence.

Protect inserted French, then apply M1 rules.

M3 | CMDR-STYLE RANKING

Shared 150-d Word2Vec over bilingual n-grams.

Score = similarity x bilingual evidence x position x n-gram penalty.

Require similarity ≥ 0.70 and evidence ≥ 0.01 ; at most two substitutions.

Protect inserted French, then apply M1 rules.

MATCHED 3,330 ROWS

M2 94.5% rows | 2.62 substitutions/row

M3 32.9% rows | 0.46 substitutions/row

Why M5 was excluded - and what M6 actually learns

M5 | DOCUMENTED, NOT HIDDEN

Unsupervised YouTube lexical mining after held-out-video exclusion.

19,828 candidates -> 114 accepted mappings.

17,777 exported rows, but only 1,821 changed; 15,956 remained identity.

Excluded because the final comparison would mainly test identity pretraining.

M6 | GOLD-TRAIN-INFORMED

865 eligible reviewed Wolof occurrences; 588 changed.

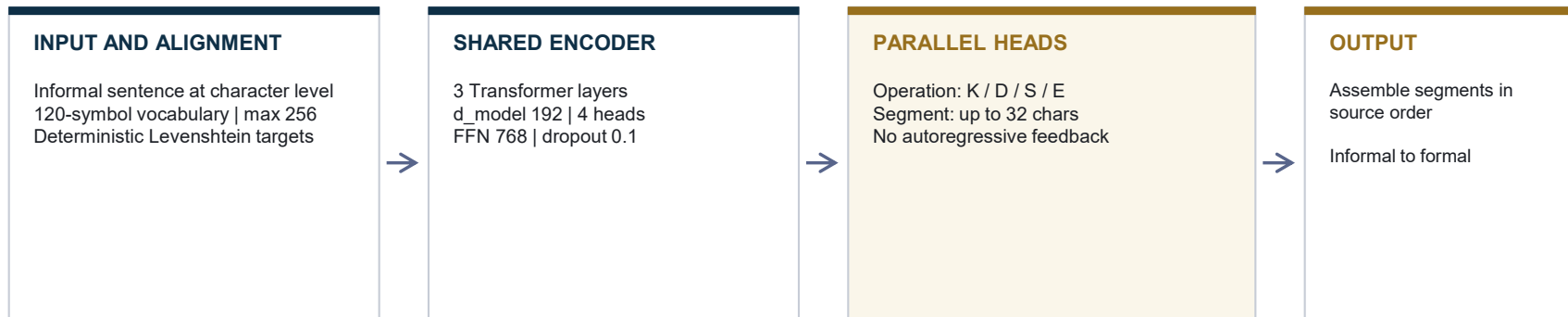
171 dominant lexical mappings

105 POS/context character templates

Source-ID-stable seed samples an empirical edit budget; candidates are weighted; overlaps and protected content are rejected.

M6 uses no Gold-dev/test annotation: supervised train-only feature extraction, not held-out leakage.

How TNT was adapted to Wolof normalization



CONTROLLED TRAINING

2.15M parameters | same vocabulary, initialization, architecture and seed
Gold-only: 20 epochs on 142 pairs
Each method: 20 synthetic epochs on 3,330 pairs -> optimizer reset -> 20 Gold epochs
Gold-dev CER selects among scheduled checkpoints; test stays sealed

Metrics, the Gold-only control, and overcorrection

WHY THESE METRICS?

CER = character edit distance / reference characters

WER = token edit distance / reference tokens

chrF = character n-gram F-score (1-6, beta=2)

Exact = complete sentence match

Plain accuracy would be dominated by KEEP and mishandle splits/merges.

STRICT CORRECTION SCORE

TP only when source span + replacement exactly match Gold.

Precision = $TP / (TP + FP)$

Recall = $TP / (TP + FN)$

Overcorrection = $FP / (TP + FP)$

Synthetic systems: 96.5-100% strict overcorrection.

WHY GOLD-ONLY = IDENTITY

It was trained for 20 epochs; the selected epoch-5 checkpoint copied all 29 test inputs. With 142 pairs and 81.1% KEEP supervision, identity was its only generalizable behaviour.

What was tried - and what should come next

ALTERNATIVES EXPLORED

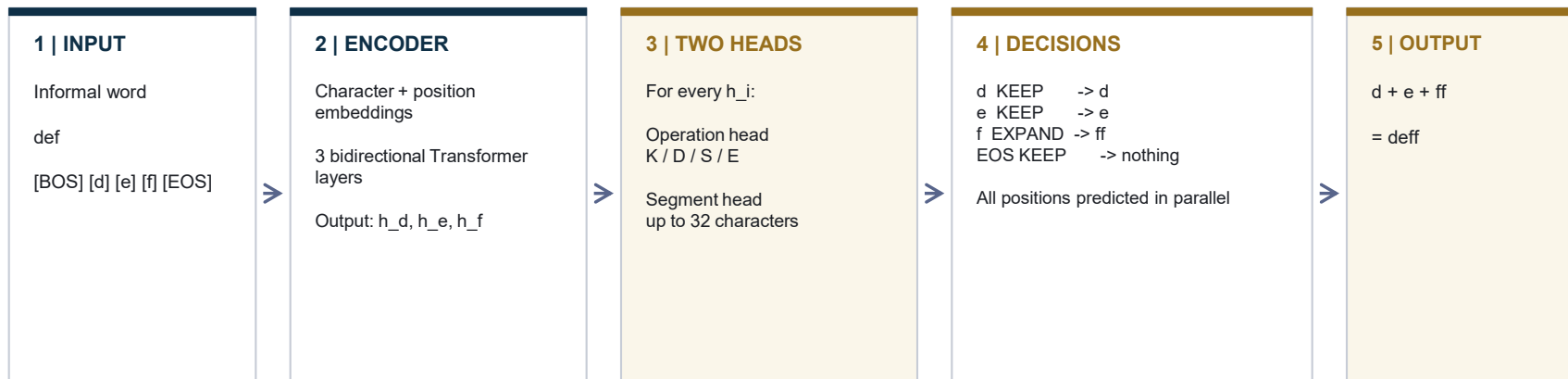
mT5-small full fine-tuning: too slow locally
LoRA: fewer trainable parameters, but full suite still costly/unstable
mBART: considered multilingual alternative
NLLB: translation-oriented
Compact autoregressive Transformer: repetition and error accumulation
Copy-aware pointer: over-copying
M5: insufficient coverage

PRIORITY NEXT STEPS

- 1 | Larger, more diverse, partly double-annotated Gold
- 2 | Balanced unchanged/light/heavy test strata
- 3 | Better calibrated generators using Gold train only
- 4 | Multiple seeds and a pretrained second evaluator
- 5 | Hybrid TNT + explicit boundary + word correction
- 6 | Retrieval, MT and LLM-query downstream tests

First strengthen authentic evaluation; then test whether the gains survive stronger models and semantic tasks.

How encoder-only TNT predicts a normalized word



WHY NO DECODER?

TNT does not freely generate a sequence. It tags each source position with an edit and a local output segment, then concatenates those segments in source order.

One alignment mechanism, several measurements

LEVENSHTEIN DYNAMIC PROGRAM

Minimum-cost path using:

match = 0
substitute = 1
delete = 1
insert = 1

Example: doff -> dof
d KEEP | o KEEP | f DELETE | f KEEP

WHERE THE ALIGNMENT IS USED

TNT training
Creates KEEP / DELETE / SUBSTITUTE / EXPAND labels for every pair.

Corpus operation statistics
The same character labels produce the Gold and synthetic operation percentages.

CER / WER
Levenshtein distance on characters or tokens.

HOW THE METRICS DIFFER

CER = (S + D + I) / reference characters
Spaces count after whitespace normalization.

WER = (S + D + I) / reference tokens
Splits and merges can create several token edits.

chrF: remove spaces; count character n-grams 1-6;
average P and R; combine with F2.

Manual labels such as phonetic, diacritic and spacing merge are separate annotations; Levenshtein only describes string operations.

Which edit count are we talking about?

A | GOLD-TRAIN KNOWLEDGE

1,315 annotated occurrences
865 eligible for M6
588 eligible changed occurrences

Learned inventory:
171 lexical mappings
105 POS/context templates

B | GENERATOR ACTIVITY

Full M6 export:
14,215 / 17,777 rows changed (80.0%)
14,809 lexical edits
22,119 template edits

Matched benchmark:
3,106 / 3,330 rows changed (93.3%)

C | FROZEN-TEST PROPOSALS

Computed from source -> model prediction using exact token-span edit blocks.

M1 68 = 2 TP + 66 FP
M2 59 = 0 TP + 59 FP
M3 68 = 1 TP + 67 FP
M4 58 = 1 TP + 57 FP
M6 57 = 2 TP + 55 FP
Gold requires 75 edits.

Why unequal? Each model changes different spans; inference has no fixed edit budget. One contiguous replacement, split or merge can count as one signature - not one per modified token.

M6 lexical mappings learned from Gold train

Dominant realizations retained from reviewed, eligible Wolof occurrences

FORMAL UNIT	INFORMAL	SUPPORT	DOM.	INTERPRETATION
am na	amna	4 / 4	1.00	spacing merge + phonetic spelling
na ma	nama	3 / 3	1.00	spacing merge + phonetic spelling
soof na	sofna	3 / 4	0.75	merge + deletion + phonetic spelling
da ñoo	dagno	2 / 2	1.00	merge + deletion + diacritic/substitution
di naa ko	dinako	2 / 2	1.00	three-token merge + deletion
moo ngi	mongui	2 / 2	1.00	merge + insertion + phonetic spelling
nak	nk	2 / 2	1.00	abbreviation
ne na	néna	2 / 3	0.67	merge + diacritic/substitution

Dominance = mapping count / all reviewed outcomes for that formal unit. Retention threshold: support ≥ 1 and dominance ≥ 0.50 .

M6 contextual rules learned from Gold train

POS and local character context turn repeated edits into reusable templates

POS	CONDITION	EDIT	EVIDENCE	EXAMPLE
VERB	final x after a	x → kh	7 / 9 · .78	wax → wakh
VERB	ë between b and g	ë → eu	6 / 6 · 1.00	bëgg → beugg
NOUN	word starts jà + m	jà → dia	6 / 6 · 1.00	jàmm → diamm
VERB	m followed by u	insert o	7 / 7 · 1.00	amul → amoul
NOUN	word ends in m	append e	4 / 4 · 1.00	koom-koom → koom-koome
VERB	word ends in l	append e	2 / 2 · 1.00	sonal → sonale
VERB	word ends in kk	delete final k	3 / 3 · 1.00	nekk → nek

Candidate rule = POS + operation + edited segment + immediate left/right context. Overlaps and protected spans are rejected during generation.

How M6 combines mappings and contextual rules

Two recorded rows from the full M6 export

EXAMPLE A · one lexical mapping + one contextual rule

Formal Areen bi amul mbër bu la mën a sonal.

Evidence amul → amoul [mapping 1/1] + sonal → sonale [VERB: append final e after l, 2/2]

Output **Areen bi amoul mbër bu la mën a sonale.**

EXAMPLE B · mapping plus four non-overlapping contextual edits

Formal Dina nu kontaan lool ci joxe dénkane yu lalu ci xam-xam wax la ba ñaata nga mënna leb te dula sonal.

Edits mënna→meuna [map] · dénkane→dénkane · xam-xam→xam-xame · wax→wakh · nga→ngua [templates]

Output **Dina nu kontaan lool ci joxe dénkane yu lalu ci xam-xame wakh la ba ñaata ngua meuna leb te dula sonal.**